and exit of certain substances. Hence, it is referred to as a semi-permeable membrane. According to this model, the plasma membrane contains a bimolecular lipid bilayer, both the surfaces are interrupted by protein molecules. The cell membranes have been visualized as mosaics of lipids and proteins.

As mentioned earlier, the cylindrical shapes and amphipathic ratio of phosphoglycerides and sphingolipids allow the bilayer formation with planar structure. This bilayer formation is associated with increased entropy owing to the hydrophobic chain of the bilayer which in turn assists in membrane assembly.

In this model, the membrane contains various molecules such as embedded protein, carbohydrate, cholesterol, and hence it is described as a mosaic pattern. The lipids are thought to be arranged primarily in a bilayer in which peripheral and integral proteins are embedded to varying degrees. The membrane proteins are not fixed within the lipid layer but are free to move laterally. The globular proteins of the membrane are of two different types: extrinsic (peripheral proteins) and intrinsic (integral proteins).

Some proteins are attached at the polar surface of the lipid, while others either partially penetrate the bilayer or span the membrane entirely to stick out on both sides. The protein usually contains chains of sugars or oligosaccharides. Some lipids present on outer surface are glycolipids. The cell membrane is more like a fluid, rather than being a rigid or solid structure. The membrane is of fluid consistency which enables the lateral movement of lipid and protein molecules within it. The proteins of the membrane are concerned with the enzymatic activities, transport of molecules and with receptor function.

Singer and Nicolson considered the lipoprotein association to be hydrophobic and fluidity of the membrane results due to hydrophobic interaction. It should be noted that phospholipids and many intrinsic proteins are amphipathic molecules, i.e., both hydrophilic and hydrophobic groups occur within the same molecule. The fluidity of lipid is evidenced based on x-ray diffraction, differential thermal analysis and electron spin resonance (ESR) techniques.

The lipid bilayer has many dynamic properties which are as follows:

1. The internal rapid motion within each lipid molecule is possible.
2. The lipid molecule might diffuse laterally
3. The transfer of lipid molecule from one side of the bilayer to the other is possible and is referred as flip-flop movement.
4. The lipid molecules might rotate rapidly as a whole about their axis.

The fluid mosaic model of cell membrane is now widely accepted as it can be applied to membranes of all types regardless of their varying characteristics and differences in lipid:protein ratio. In fact this model can account for the molecular organisation and ultrastructure of membranes in-terms of their chemical composition.

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